



DEALER BEST PRACTICE SERIES

Hydraulic Shop Stand for Lubrication of Suspension Cylinders During Assembly

Maintenance and Repair	Application	Component Life Management	Component Renewal (CRC)	MARC Management
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1.0 Introduction

During the rebuild of suspension cylinders, adding oil to the cylinder in the final step is time consuming. The processes in most of the Disassembly / Assembly guides are to mount the cylinder in a vertical position, add the oil, and then add install the head of the cylinder.

Today, the hydraulic shop has fabricated a stand for the lubrication of the cylinder.

Before this Best Practice, the cylinder was placed vertically using a jib crane all the time, in that position the cylinder was filled with oil. The average time of this operation was two hours. At the present time, it only takes ten minutes.

D & A Guide Process

17. Fill housing (6) with 73.4 L (19.4 US gal) of Tooling (L) .

[View Image](#)



Illustration 15

g01933413

18. Apply Tooling (G) onto ring (4) and O-ring seal (5) . Install O-ring seal (5) and ring (4) .

[View Image](#)

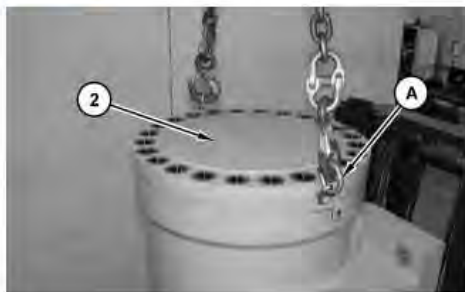


Illustration 16

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2.0 Best Practice Description

A horizontal stand has been designed to hold the suspension cylinder. The stand has a tray in the base to make sure no hydraulic oil can spill on the shop floor.

The use of the stand reduces the time required by 90% and the bay jib crane is only required for a few minutes to mount the cylinder instead of the normal 2 hours. The crane can then be used for other tasks.

See photo 1.

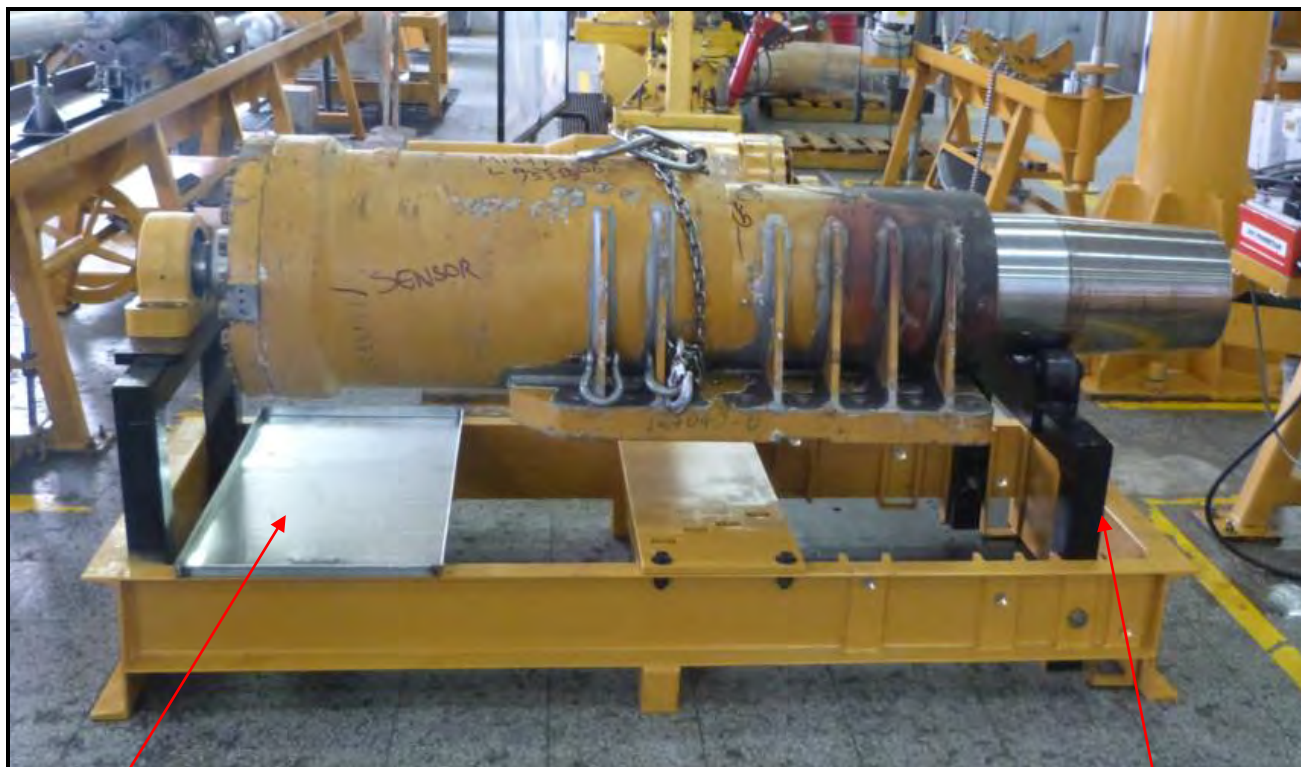


Photo 1: Suspension Cylinder Stand

Drip Pan

Adjustable
Support

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3.0 Implementation Steps

- Fabricate a lubrication stand. Prints are provided below.
- The process
 1. Assemble Cylinder is placed on the stand using the jib crane. Photo 2



Photo 2. Cylinder placed

Adjustable Support –

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2. Rotate the cylinder 90 degrees. Photo 3 below.



Photo 3. Direction of rotation

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3. Install stand lock bar in order to keep the cylinder in position.



Photo 4. Lock bar

4. Fill with oil using an oil pump.

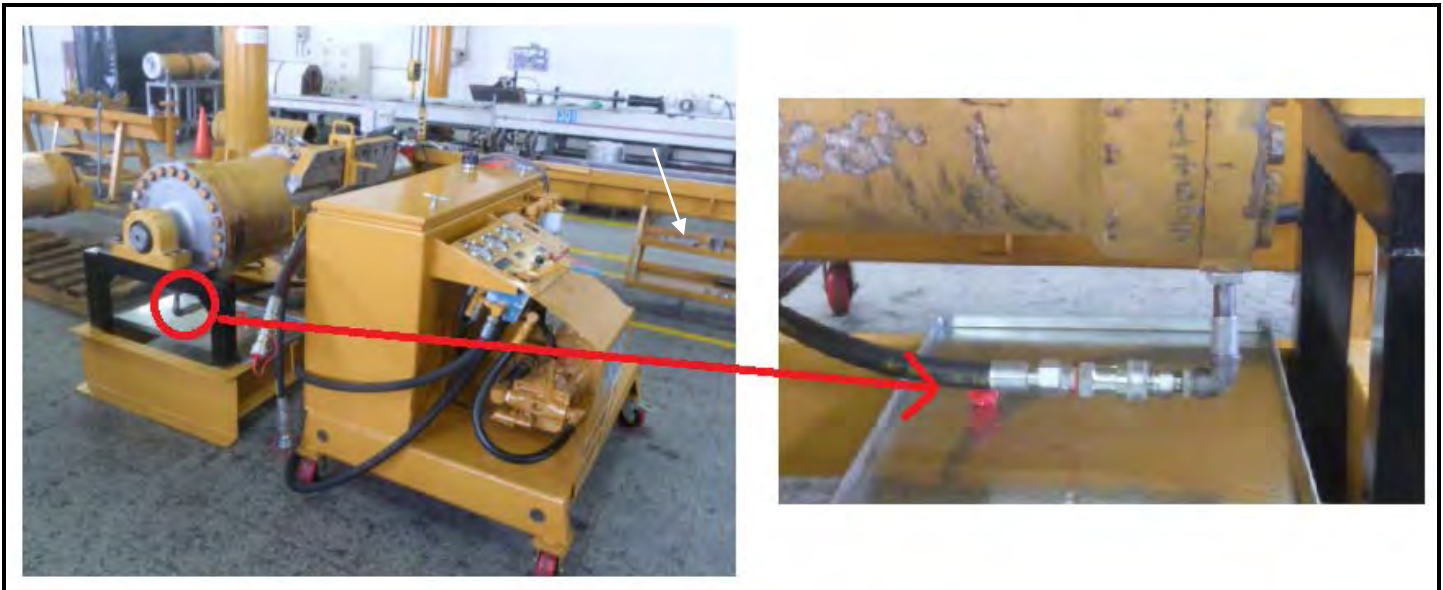


Photo 5. Fill with oil

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4.0 Benefits

- Reduce lubrication time by 90%.
- Before this Best Practice, the jib crane was used all the time during lubrication process. Now the jib crane is just used for eight minutes approximately.
- Improve ergonomics and safe handling.

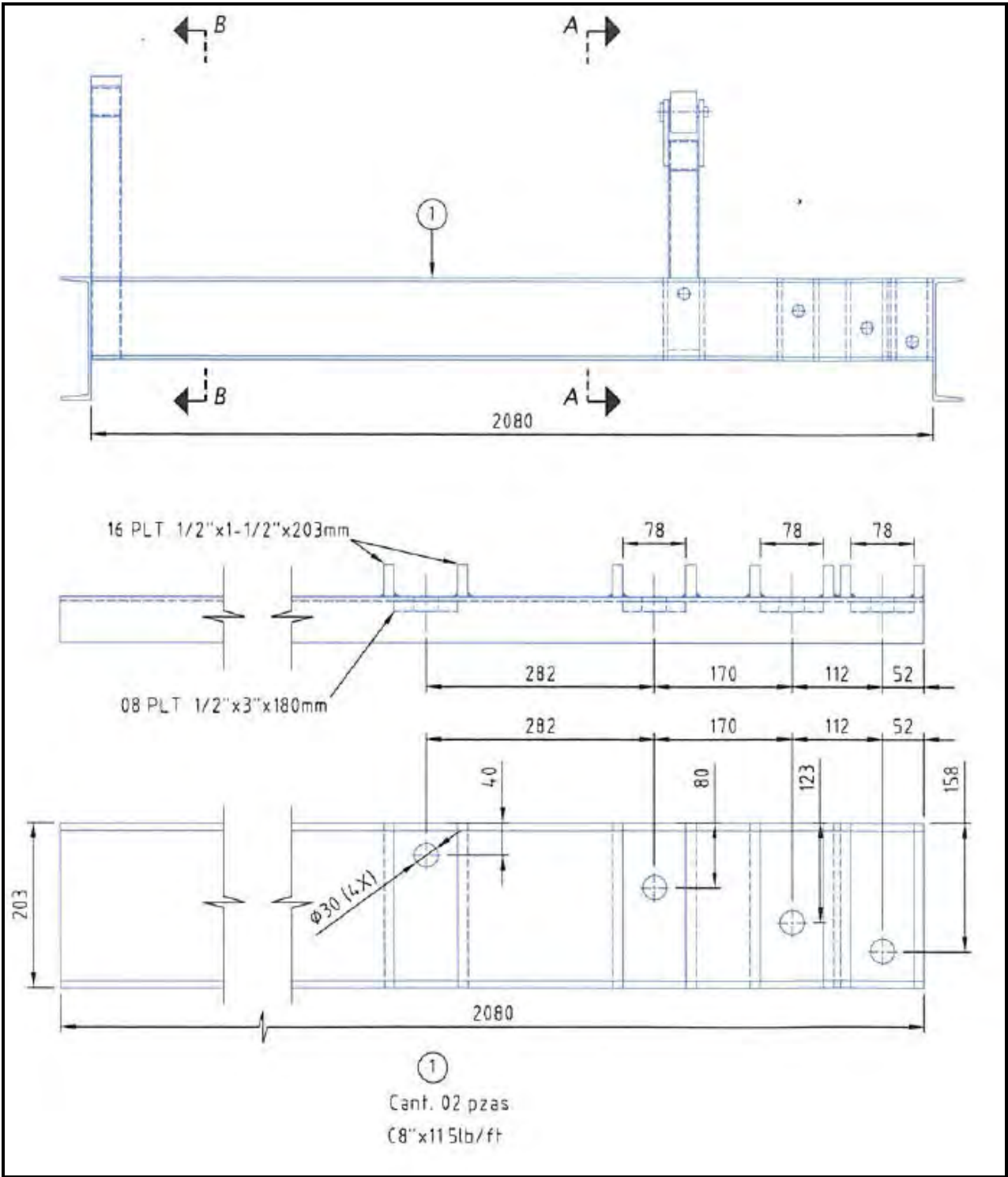
5.0 Resources Required

- Fabrication Workshop.
- The average cost of this device is \$ 1,500 USD (based on \$50/hour labor rate)

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6.0 Supporting Attachments / References



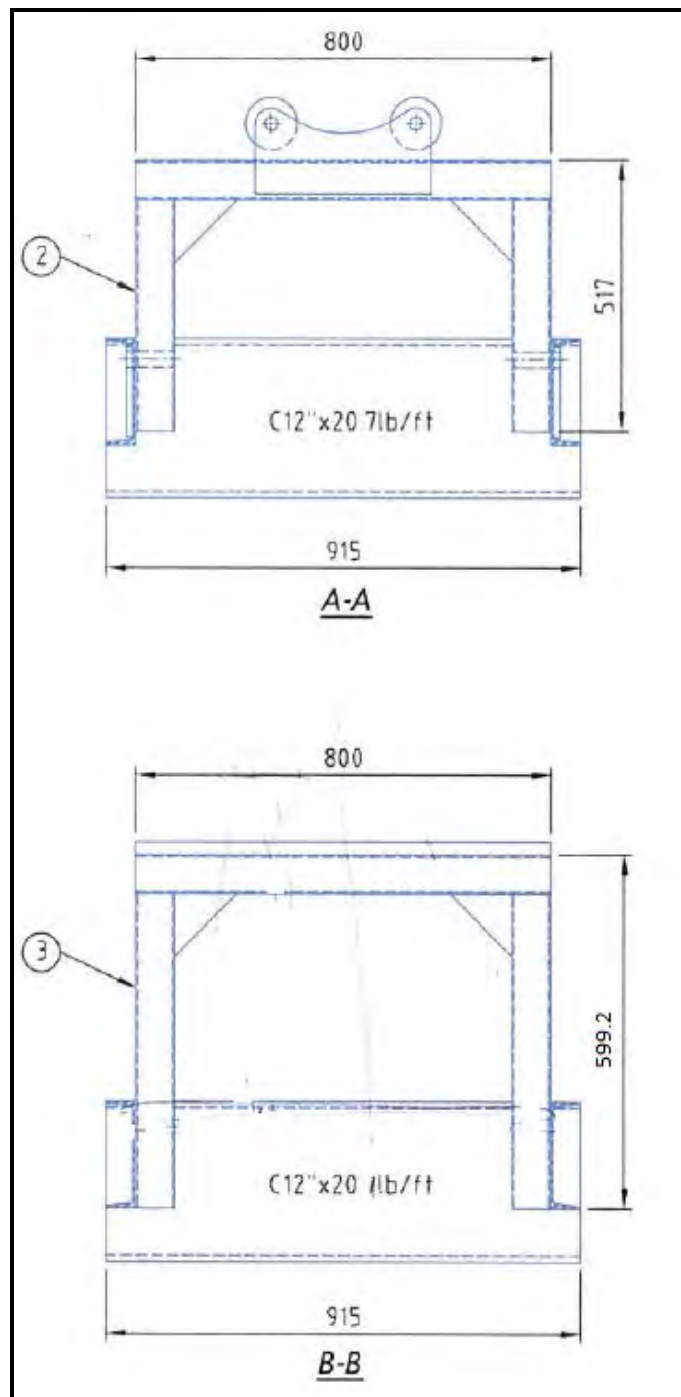
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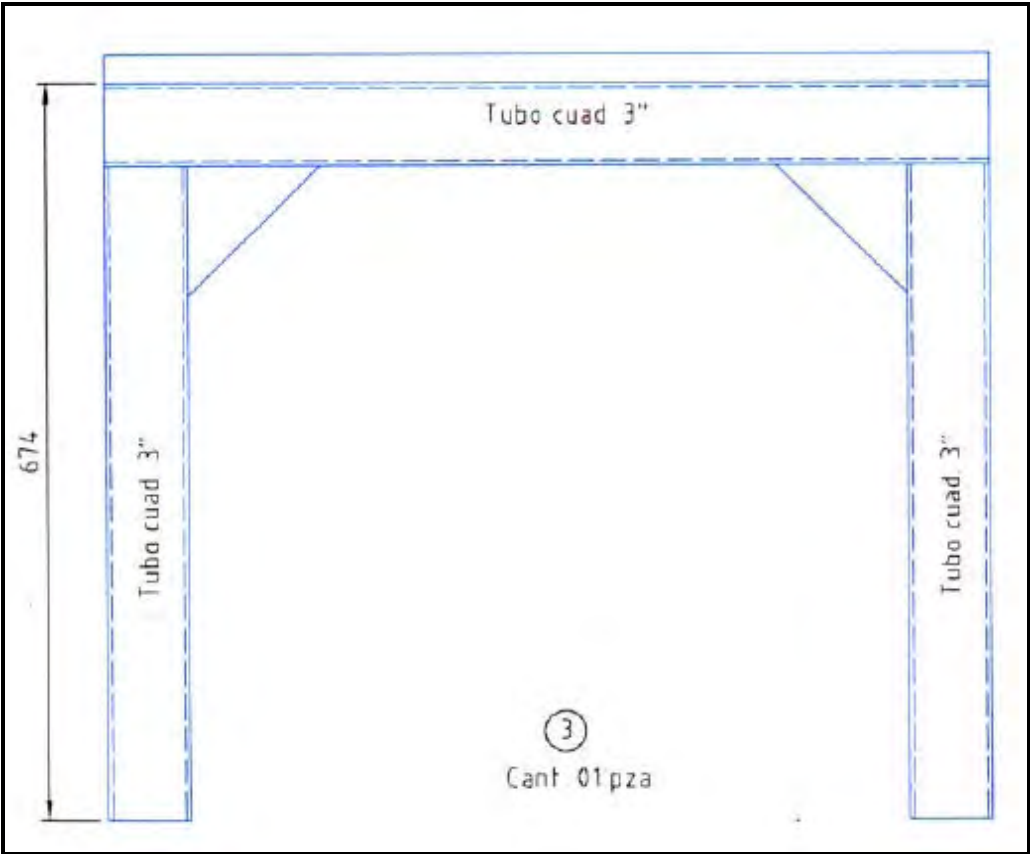
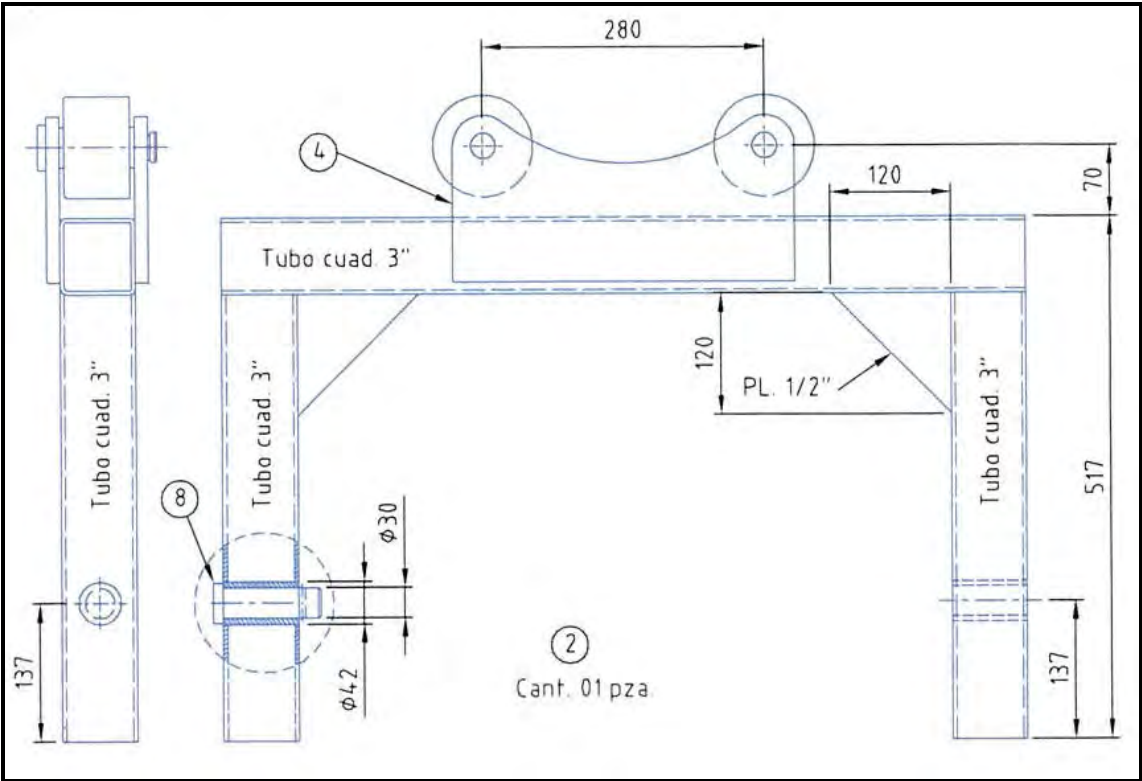
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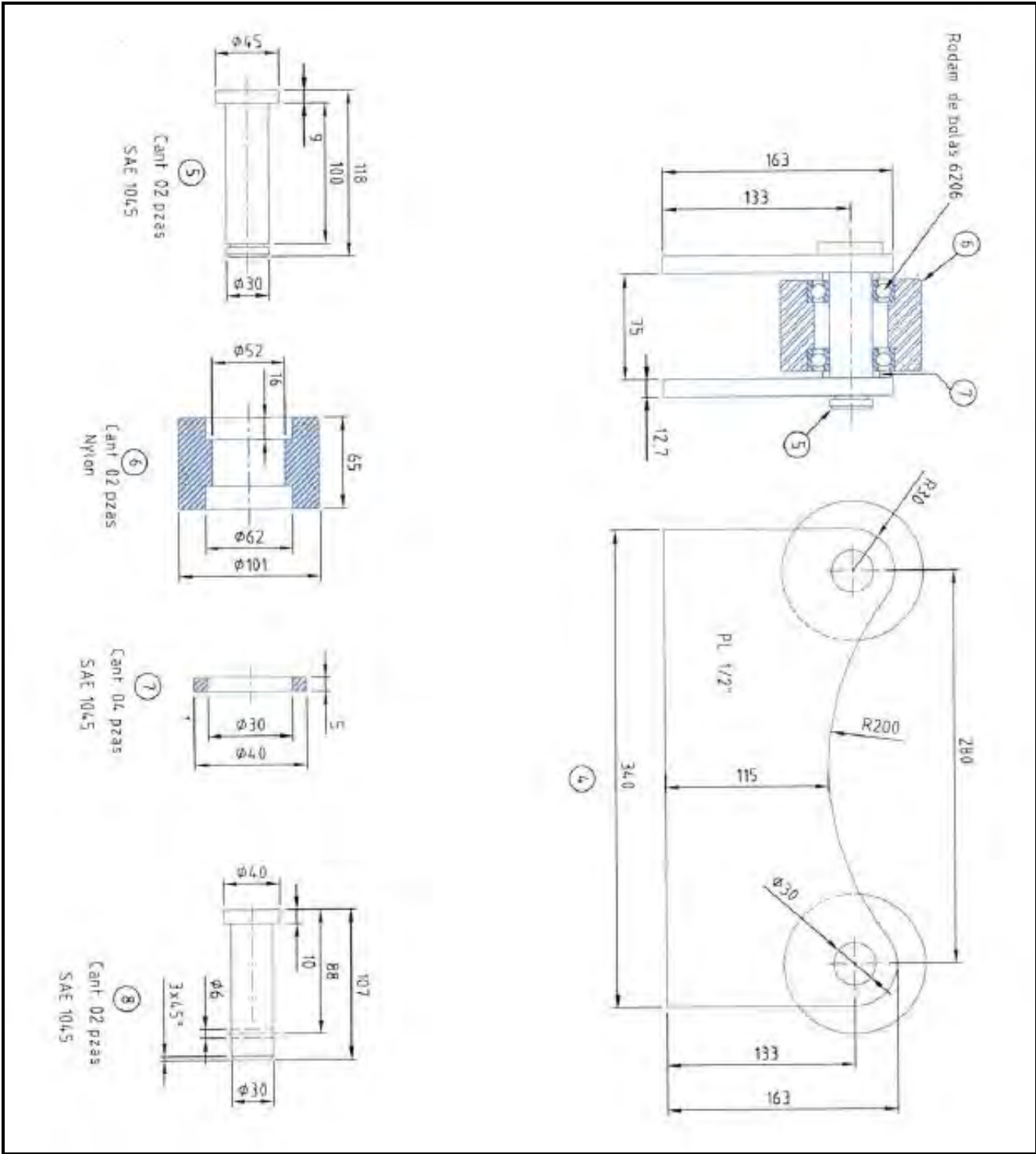
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7.0 Related Best Practices

None

8.0 Acknowledgements

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